

$$\operatorname{Re}\langle u, v \rangle = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

Najprej razpisimo

$$\|u+v\|^2 = \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle$$

$$\|u-v\|^2 = \langle u-v, u-v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle$$

$$\|u+v\|^2 - \|u-v\|^2 = 2\langle u, v \rangle + 2\langle v, u \rangle : (u, u \text{ a } v, v \text{ se odštejeta})$$

v kompleksnem p. velja $\langle v, u \rangle = \overline{\langle u, v \rangle}$

$$\|u+v\|^2 - \|u-v\|^2 = 2(\langle u, v \rangle + \overline{\langle u, v \rangle})$$

• prebrskamo (dobimo $z + \bar{z} = 2\operatorname{Re}(z)$)

$$\|u+v\|^2 - \|u-v\|^2 = 2(2\operatorname{Re}\langle u, v \rangle) = 4\operatorname{Re}\langle u, v \rangle$$

$$\operatorname{Re}\langle u, v \rangle = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\operatorname{Im}\langle u, v \rangle = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$

$$\bar{i} = -i$$

$$\|u-iv\|^2 = \langle u-iv, u-iv \rangle = \langle u, u \rangle - \langle u, iv \rangle - \langle iv, u \rangle + \langle iv, iv \rangle = \langle u, u \rangle - i\langle u, v \rangle - i\langle v, u \rangle + i^2\langle v, v \rangle$$

$$= \|u\|^2 + i\langle u, v \rangle - (i\langle v, u \rangle) + \|v\|^2$$

$$\|u+iv\|^2 = \langle u+iv, u+iv \rangle = \langle u, u \rangle + \langle u, iv \rangle + \langle iv, u \rangle + \langle iv, iv \rangle = \|u\|^2 + i\langle u, v \rangle + i\langle v, u \rangle + \|v\|^2$$

(sahaj iselj brok abo v prebrskamo)

$$\|u-iv\|^2 - \|u+iv\|^2 = (i\langle u, v \rangle - i\langle v, u \rangle) - (-i\langle u, v \rangle + i\langle v, u \rangle) =$$

$$= 2i\langle u, v \rangle - 2i\langle v, u \rangle = 2i(\langle u, v \rangle - \overline{\langle u, v \rangle})$$

$$z - \bar{z} = 2i\operatorname{Im}(z)$$

$$2i(2i\operatorname{Im}\langle u, v \rangle) = 4i^2\operatorname{Im}\langle u, v \rangle = -4\operatorname{Im}\langle u, v \rangle$$

$$\operatorname{Im}\langle u, v \rangle = \frac{1}{4} (\|u-iv\|^2 - \|u+iv\|^2)$$